

12 (a) A medical clinic uses objects of the class `Patient` to assign a priority and a doctor to a patient. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
PatientID	STRING	Unique identifier of the patient
Name	STRING	Patient's full name, surname first
DoctorID	STRING	ID of doctor administering treatment

Treatment is prioritised with a numeric scale of 1 to 5.

Complete the class diagram for `Patient`, to include:

- attribute and data type for the date of birth
- attribute and data type for the priority
- methods to assign the patient ID, priority and doctor ID
- methods to return the patient ID, patient date of birth and the priority.

Patient	
PatientID	: STRING
Name	: STRING
.....	:
.....	:
DoctorID	: STRING

.....

SetName(FullName : STRING)

SetDateOfBirth(DOB : DATE)

.....

.....

.....

GetName()

.....

.....

GetDoctorID()

[5]

(b) (i) Identify the object-oriented programming (OOP) term described as ‘an occurrence of an object’.

..... [1]

(ii) Describe what is meant by the OOP term **polymorphism**.

.....

.....

.....

..... [2]

13 The pseudocode algorithm below uses random file access to copy 50 records from a live file `CurrentResults.dat` to a stored file `StoredResults.dat` one record at a time. It uses the user-defined type `StudentResult`.

```
TYPE StudentResult
  DECLARE LastName : STRING
  DECLARE FirstName : STRING
  DECLARE ExamGrade : STRING
ENDTYPE
```

If any grades are missing in `CurrentResults.dat`, the text "Missing grade" is added to the `ExamGrade` field in `StoredResults.dat`

Complete this file handling pseudocode algorithm.

```
DECLARE Grade : StudentResult
DECLARE Position : INTEGER

.....

OPENFILE "StoredResults.dat" FOR RANDOM

.....

  SEEK "CurrentResults.dat", Position
  GETRECORD "CurrentResults.dat", Grade
  IF Grade.ExamGrade = "" THEN

    .....

  ENDIF

  .....

  .....

NEXT Position
CLOSEFILE "CurrentResults.dat"
CLOSEFILE "StoredResults.dat"
```

[5]